Visual Analysis of Child-Adult Interactive Behaviors in Video Sequences

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***Abstract*—Kids activity means a lot to their parents, and in the analysis of the activities, video retrieval has played an important role. In this paper, we propose an effective approach for the retrieval of the kid's activities from home videos. The video sequences are taken from our test-bed environment that is designed in the form of a smart home, and feature various types of child-adult interactions. We present a novel retrieval method with two steps, first using spatio-temporal matching to obtain a coarse result, And then we propose a method to learn dominant child-adult interactive behaviors based on a sequence of home videos. Based on these dominant behaviors, we get rid of some false retrieval and obtain fine result. We implement and test our methodology on a newly-introduced dataset containing several types of kid's activities, and the retrieval result shows its potential application in the video analysis demain, it can find out most of the video clips relevant to the query one.**

I. Introduction

With the increasing proliferation of digital video contents, efficient techniques for analysis, indexing and retrieval of videos have attracted many researchers attention. Usually, the first step for video analysis is to segment a video into elementary shots, each comprising a continuous in time and space. And then analyze each segments separately. However enormous the data is, finding interesting information from it is very difficult and arduous, and people neither have time nor the energy to browse large amounts of data to find what they need. So how to automatically retrieve the video information we needed is an interesting and challenging problem. In the practical applications, take the assessment of the autism by analysis of child's behavior for example, there will be a large amount of home videos taken by parents or video sequences taken in controlled settings where a child engages in a social game with a parent or another familiar adult (see Figure 1), these visual records can often provide clues to the child's behavior which can assist a specialist to determine if any observed behavior pattern may necessitate further examination. Such analysis can lead to early detection of developmental issues such as autism or attention disorder. For the purpose of finding out the desired segments that are useful for the diagnosis and treatment of children with autism, we need to do some retrieval to help us find the clips we really need. The problem of extracting the traits of the child's behavior from the video dataset is the problem we want to solve in this paper.

Fig. 1. Child-adult interactive behaviors. These videos are taken form Internet.

In this paper, we propose a novel spatio-temporal matching method to retrieve video clips from the dataset. And also, for the refinement stage, we propose a method to learn dominant child-adult interactive behaviors based on a sequence of home video. For some kind of retrieval, we take the context information into consideration, and use a context modeling for the further refinement of the retrieval results. Our method is motivated by the success of human behavior recognition via sparse spatial-temporal features, and the effectiveness of "bag of visual words" model in the representation of the human activities.

To summarize, we make three main contributions in this work:

- First, we introduced a new video retrieval method based on the spatio-temporal features to find out the video clips relevant to the query one.
- Second, we propose a method to learn dominant child-adult interactive behaviors based on a sequence of home video.
- A publicly-available video dataset of child-adult interactive behaviors are taken in our test-bed environment in the form of a smart home, featuring various types of child-adult interactions.

The remainder of this paper is organized as follows. In Section II we discuss related work. We describe our methods in Section III. In Section IV we present the result of our

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algorithm with the child-adult interactive behavior dataset taken by ourselves. We conclude in Section V.

II. Related Work

A. Content Based Video Retrieval

In the domain of the video retrieval, content-based search and retrieval has become a major focus. In this area, how to model the video content is one of the most important problems in video retrieval, and a large amount of work has addressed this problem. In all, video content can be modeled at different levels [16], such as raw data, low-level visual content and semantic content. Raw data refers to some elementary video units as well as some general video attributes. While low-level visual content is based on visual features such as colour, shapes, textures etc. And another important level, Semantic content is consist of high-level concepts that can be presented through different visual presentations. For the complexity of this level, its extraction process can be difficult for the need of specific domain knowledge or user interaction. While little efforts made in the raw data level, there are much more work related to visual content level [2]. These retrieval systems allow a user to make queries based on visual image content, such as VisualSEEk [19], Photobook [15], as well as Virage video engine [7] and VideoQ [4] that related to the videos. These approaches above are mainly applying similarity measurement on the extracted key-frame based on the low-level features. The approaches discussed above are query by example approaches, these are suitable if a user has a similar image at hand, but if the image is taken from a different angle or has a different scale, the match in terms of the feature metrics may yield poor results, and they would not perform well. Another insight into the video content modeling is to model the semantic content, although it is far more difficult than the low-level one. For a video is just a temporal sequence of pixel regions, it does not have any direct relation to its semantic content, thus it's difficult to explore them from the raw data. One way to model the video content using semantic is to use free text manual annotation. Some work [1][5] also introduce objects and events, as well as their relations that should be annotated. Another way to model the video entities is to use the spatio-temporal relations [8][10], these spatio-temporal relationships between objects in the video are the high-level concepts, and they are an important cue in content-based retrieval.

B. Activity Recognition Based on Spatio-Temporal Words

Recognizing activities through spatio-temporal words is an emerging area that causes many researchers concern. These features are robust to noise, due to small camera shake and sudden changes in lighting conditions. Laptev et al [9] proposed a space-time interest point detector based on the idea of the Harris corner detector, by using these extracted feature points to represent actions, then a SVM classifier is applied to recognize human actions. They show promising results in human behavior recognition, and mainly focus on the walking people in the scene. Dollàr et al. [6] propose an alternative approach to detect sparse space-time interest points, the detected points is called "cuboids", and it is more probably pay attention to the periodic motions. In the work, he represented an activity as a bag-of-features and applied SVM to classify videos containing each activity. Moreover, these extracted local space-time patches have been proven useful to provide semantic meaning of video events by providing a compact and abstract representation of patterns. Niebles et al. [14] use the feature detector proposed by Dollar, and use the probabilistic latent semantic analysis (pLSA) to learn the action categories, and also they applied a spatial clustering to localize actions. However, their work ignores structural (or geometrical) information of the feature points. While in [18] they propose a video matching approach, combine the spatio-temporal relationships into the matching kernel, and compute similarity between two sets of features extracted from different videos. This match can be suitable for detecting many complex non-periodic activities. Ping Wang et al. [20] use the spatio-temporal words to represent a kind of game called "social games", and developed a method to extract these quasi-periodic pattern from videos. This work provides us a new way for analyzing video contents by using spatio-temporal features.

C. Finding Dominant Behaviors from Video Sequences

Observing the behavior of a child when interacting with adults on playing with toys or in games such as peek-a-boo can offer hints to the developmental behavior of the child. Home videos taken by parents or video sequences taken in controlled settings where a child engages in a social game with a parent or another familiar adult (see Figure 1) can often provide clues to the child's behavior which can assist a specialist to determine if any observed behavior pattern may necessitate further examination. Such analysis can lead to early detection of developmental issues such as autism or attention disorder. The problem of analyzing the video to extract the traits of the child's behavior exhibits many challenges including variations in the visual data caused by camera placement, camera motion, lighting, and subject's clothing and appearance. The current practice in such studies involves an extremely labor-intensive analysis of video recordings of the child during such interactions [20].

D. Object Recognition with SIFT Features

The SIFT is an approach for extracting distinctive invariant features from images. It was first proposed by Lowe [12]. It is widely used due to its invariance to scale, orientation, and affine distortion, and partially due to its invariance to illumination changes. Owing to its discriminative power as a feature, a lot of work have been addressed on the recognition using SIFT features. Mohamed Aly[13] use the SIFT features for the recognition process, he computed the SIFT matches between the given face image and all faces in the database. Those with largest number of matching points is considered as the nearest face, and is used for the classification of the new face. Faraj Alhwarin [3] proposed an improved SIFT features for the object recognition. While SIFT-based methods are expected to perform better for objects with rich texture information, for there will

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be sufficient number of keypoints to be extracted. On the other hand, they also require sophisticated indexing and matching algorithms for effective object recognition [11][17].

III. Methods

In this work, we mainly focus on the retrieval of the kid's activities. Most part of the activity in the dataset is kid's game playing, it contains playing balls, doing cookings, reading books, etc. Our work is to retrieve one of these activities. Given an input video segment which contains a kid's activity, we can search from the video dataset, and find out the video segment which contains the activity similar to the input video. To do this, we use a spatio-temporal words match methods, which measures the similarity between video segments. We use spatio-temporal features to extract the points that indicate a periodic motion or a spatio-temporal corners, those interesting points play an important part in analyzing the human motion. Afterwards, we make use of a spatio-temporal element called "cuboids" which was proposed by Piotr Dollar [6] to describe each feature points and get a feature vector, then clustering them, to form the spatio-temporal words. In this way, a video segment can be represented as a histogram of its spatio-temporal words frequency. Then match the similarity between the input video and the video segment in the dataset, when the similarity result satisfy a certain threshold, we find out it and make it the candidate for the further selection. Base on the work above, when we get a series of video segment which similar to the input one, also some unrelated video segment are be found as a result. Thus a problem comes, that is how can we discriminate the video we want from the all the searching result. To solve this problem, we propose three methods: first, as the search procedure is a matching procedure which measures the similarity between video segments, thus we can rank the similarity value and this ranking can be a clue to indicate which segment may more "similar" with the input one. Except for this methods, we also find out the dominant child-adult interactive behaviors from video sequences, as the query input is one of these dominant activities, we use this property as a clue to eliminate some unrelated retrieval result. Thirdly, for some kind of activities , we may consider the objects which the kid is playing as another important clue to indicate which game the kid is playing. Thus through these three methods, we can improve the performance of our retrieval systems, and get desirable results.

A. Spatio-Temporal Matching of the Video Segment

1) Video segments: The input video is selected from the dataset, it contains a kid's activity which lasts for nearly 10 seconds. The selected segments contains a periodic motion of the kid's game playing. In the dataset video, it contains many children's games, some related to the input, while other not. We manually segmented the dataset into clips of length t_win using a sliding window. Each segment contains almost 200 frames, and without overlapping with adjacent segments, so that we can find all of the video segments similar to the input one, also avoids unnecessary computations. We use $t_win = 13s$ on average in our experiments. And each segment is processed independently.

Fig. 2. The result of feature detection

2) Spatio-temporal features: In the course of feature extraction, we use spatio-temporal feature extraction methods proposed by Piotr Dollár [6]. In this work we adopt the concept "cuboids" to describe the feature point. First, the response function of the feature has the form , where $g(x, y; \sigma)$ is the 2D Gaussian smoothing kernel, applied only along the spatial dimensions, and h_{ev} and h_{od} are a quadrature pair of 1D Gabor filters applied temporally. h_{ev} and h_{od} are defined as $h_{ev}(t; \tau, \omega) = -\cos(2\pi t\omega)e^{-t^2/\tau^2}$ and $h_{od}(t; \tau, \omega) = -\sin(2\pi t\omega)e^{-t^2/\tau^2}$, for simplicity let $w = 4/\tau$, and the σ and the τ, corresponding roughly to the spatial and temporal scale of the detector. In this experiment, we let spatio scale $\sigma = 2$, and temporal scale $\tau = 3$. All the points with local maxima response value was retained, and regarded as the feature point. The feature extracted is given more attention to the periodic motion and some spatio-temporal corners. Then a cuboids was extracted around the feature point and it contains the spatio-temporally windowed pixel values, this can make a contribution to describe features. Thus, many cuboids descriptors can be used, in this experiment, we use descriptors called "GRAD", that is a patch descriptor based on histogrammed gradient, which is fairly robust to small perturbations of the image. It involves flattening the cuboid into a vector, and then reduce the dimensionality of the descriptors with PCA. In this work, the final dimension of the feature descriptors is set to 100. Figure 2. shows the result of detection.

3) Spatio-temporal words: After we get the feature descriptors, then each interest feature point is represented by a feature vector f_p, each f_p has 100 dimensions, then we build spatio-temporal words by applying K-means clustering algorithm to the feature vectors $\{f_p\}$, we set the cluster number to be 30 as default. Each of the feature point is then assigned to a closest visual word. Figure 3 shows this clustering step.

4) Video descriptor: Each video segment is represented by a histogram of its visual words it contains. By this way, we discard the spatial information and temporal information of the feature points. The results show its simplicity to do the retrieval more efficiently. Figure 4 shows the histogram of the video.

5) Video matching: This step is to compute the similarity between the video segments, after we get a descriptor for each video segment. Then the distance between video segments can be calculated by using the Euclidean distance. We sort the distance by ascending order, and choose the ones that rank top 15 or 20 of the distance as the main retrieval result. In this way, the video segments founded are more "similar" to the

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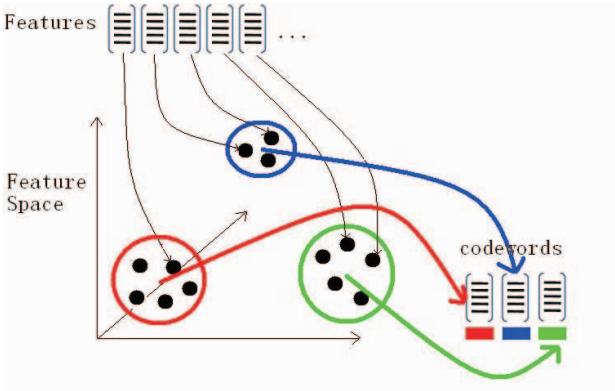

Fig. 3. K-means clustering of descriptors

Fig. 4. Video descriptors

input one.

The first step accounts for the coarse retrieval of the kid's activity, for the further processing, we need to get a more precise results of the retrieval, eliminate some unrelated results, and to get a fine retrieval of the activity.

B. Finding Dominant Child-Adult Interactive Behaviors

Based on the work above, the input video that records a set of child-adult interactive behavior types is segmented into clips using a sliding window. Then the space-time interest point detector proposed by Laptev [9] is applied to each video segment to get the interest points. While spatio-temporal neighborhoods of local features contain information about the motion and the spatial appearance of events in image sequences. To capture this information, we get the feature descriptors. K-means clustering of descriptors j in the clips gives a vocabulary of primitive events h_i . The numbers of features with labels hi in a particular sequence define a feature histogram $H = (c_1, ..., c_n)$. We use such histograms as the clips' representations. Every clip is represented by a histogram of the visual words that it contains. We then assign a label to each clip by applying k-means clustering to the histogram representations of clips (see Figure 5). By counting the amount of each label, we can get the label which has the largest count. Therefore, child-adult interactive behavior in clips which have this label should be the Dominant child-adult interactive behavior in the input video.

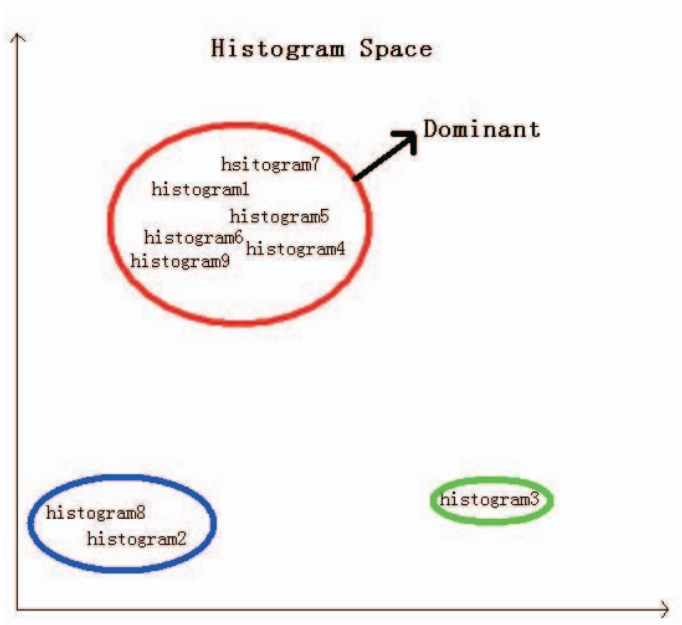

Fig. 5. K-means clustering to the histogram representations of clips

C. Context Modeling for Refinement of the Video Retrieval

For some type of kid's activities, the kid is playing with a certain object, such as toys, balls, etc. Inspired by this, we can use the context information as an important clue to infer the activity which the kid is performing. In this section, we proposed a context modeling method to make the retrieval more precise. Here, we detect the object that the kid is playing with, and use this object as a clue to infer the activities. Going through this step, the experiment results show this method can improve our retrieval performance in some extent. For the context modeling, the key point is to recognize the object which the kid plays with, as there exist many methods to solve it, we adopt a simple matching scheme with the SIFT features to find the object in the video clips, and use the matching results to determine whether the object is contained in the video clip. If the object is occurred in the video clip, it is more probably that the kid is playing some kind of activity related with the object. That is another clue to improve the retrieval performance.

1) Sift matching: We get an image patch sample which contains the object as another query input, and compute the SIFT keypoints in it, then when the SIFT features are obtained, we compute the SIFT matches between the input patch and the each frame in the video. Afterwards count the number of the matches. For some of the pairs the number of the matches counts to 0 or 1, although it contains the object in the frame. While some other image pairs have more matches. Thus we just consider the pairs which the match number is above a threshold. In this experiment, we set it to 3, for most of the matches counts to 3 or more, this is a empirical value, and can change it due to the different object.

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Fig. 7. Activities used in the retrieval stage

IV. Experiment

We present three different kind of experiment to evaluate the performance of our methods for retrieving kid's activities from the video sequence. First, we experiment on the activity retrieval, this stage is the coarse retrieval stage, for we just applied the spatio-temporal matching algorithm. In the second experiment, we find the dominant child-adult interactive behaviors from the video dataset, and use this dominant behaviors to refine our retrieval results. Third, we try our context modeling method in some kind of activities for the refinement of the retrieval process.

A. Datasets

In order to test our system, we have constructed a new dataset, which contains more than five typical kinds of children's games, including playing balls, reading books, cooking, playing toys, play with pans. In the dataset, some kind of activity lasts for a long time, while others not. For this reason, in this experiment, we mainly focus on the retrieval of the activity which lasts for a long time and accounts for most of the dataset, that is playing balls, reading books, do cookings. The input video segment is selected from the dataset, each input contains one kind of the three activities, and lasts for almost 10 seconds, our task is to search through the dataset, and choose the video segment which contains the same or similar activities. Also, to make the comparison more easily, we segment the dataset video into clips using a sliding window, resulting in each clip having a length of 200 frames, nearly 13 seconds each one. As a result, one kind of activity may be separated into several video clips, and all of these will be found out as the same activity searching result. Figure 6 shows some snapshots of the dataset.

B. Activity Retrieval

In the coarse retrieval stage, we use each activity video sample as a query, and test the performance of the retrieval methods. And for the query sample, only three kind of activities are used, they are ballplaying, reading and cooking. Figure 7. are the activities used for retrieval.

We take the ballplaying video sample for example. By applying feature detection to the input clips, and the spatio-temporal words are built by applying k-means clustering to all feature points, The number of clusters is set to 30 by hand. We then represent the input clip by the histogram representation of the spatio-temporal words with the dimension of 100. Then the dataset video is segmented into 120 clips, each contains 200 frames, lasts for nearly 13 seconds. Each clip is processed as above, and get a video descriptor as well. Then an similarity measure was defined. By using Euclidean distance, we can get a distance value between the input video and each dataset video segment, these distance value plays a key role in the matching stage. For simplicity, we sort these distance value, and use this order as the similarity measurement. Those with smaller value may be considered as the ones that more similar to the input sample. In order to test our algorithm in an effective and efficient way, we select the retrieval result from the top of the matching results, in actual use ,we choose the ones that rank the top 20 to make the comparison more easily. Table I. shows our coarse retrieval results. In the retrieval of the ballplaying games, we also find out some unrelated ones, including cooking activity, playing toys, random walking (It was not in the defined activities) and so on. That is because our spatio-temporal matching approach cannot be fully competent in discriminating human behaviors. In detail, when we get the video descriptors by histograming of the visual words it contains, we discard the spatio and temporal information between feature points, thus in this way, some unrelated activities that contains the similar feature information may contribute to the retrieval results. That is why single retrieval step cannot get fine results. Figure 8. gives a comparison of the retrieval performance for the different games that used for retrieval. From the figure, we can see that our average accuracy rate yields a precision close to the value $P_{uniform}$. We can also find that when at 10% - 40% recall, the precisions of all the activities are nearly 50%, this is partly due to the performance of the matching methods. And it is

TABLE I
Summary of videos for the coarse retrieval experiment. Clips were segmented into 200 frame windows and manually labeled, the number of the clips is 120, we choose the similarity that rank the top 20

Game	#Related	#Unrelated	$P_{uniform}$
Ball playing	8	12	40%
Cooking	10	10	50%
Reading	8	12	40%

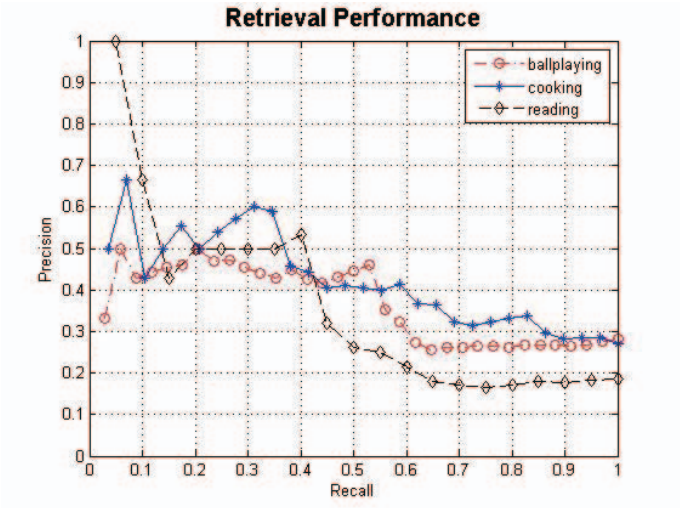

Fig. 8. Precision(Y axis)-Recall(X axis) curves

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Fig. 6. Child-adult interactive behavior dataset

Fig. 9. Snapshots of input video

TABLE II
THE AMOUNT OF 10 TYPES LABEL FOR 120 CLIPS

Label	0	1	2	3	4
Amount	0	29	0	10	4
Label	5	6	7	8	9
Amount	2	2	72	0	1

the main reason why the precision cannot reach 100% when recall approaches 0. Figure 9. shows the input of our ballplaying query. Figure 10. shows the snapshots of the unrelated results. While figure 11. shows the related ones.

C. Finding Dominant Child-Adult Interactive Behaviors

In this experiment,the dataset video is segmented into 120 clips with a sliding window as processed above, each clip has a length of 200 frames. The space-time interest point detector proposed by Laptev [9] is applied to each video segment. To capture the information of local features, we compute spatio-temporal jets at the center of each feature using normalized derivatives. Spatio-temporal words are built by applying K-means clustering to all spatio-temporal jets. Each interest point is assigned to a closest visual word. We then assign a label to each clip by applying k-means clustering to the histogram representations of clips. In this experiment, since there are about 10 kinds of child-adult interactive behaviors in the video sequences, the number of clusters to these histograms is set to 10 by hand. The counts of the 10 label types for the 120 clip histograms are listed in Table II.

Clips with the label 7 (label 7 has the largest count of 72) correspond to 7 types of child-adult interactive behaviors: pointing to somewhere, pushing and pulling a toy, and playing with cooking utensils (see Figure 12) and so on. Those child-adut interactive behaviors are the most common ones among the interactions appeared in the video. Within these 7 activities, it contains the three queried activities. For this reason, we could use these founded dominant behavior as a indicator to eliminate some unrelated retrieval results.

Fig. 12. The three types of dominant child-adult interactive behaviors among the interactions appeared in the video. (a) is the behavior that the child talks to adults, (b) is the behavior that the child pushs and pulls a toy, (c) is the behavior that the child plays with cooking utensils.

TABLE III
SUMMARY OF VIDEOS FOR THE REFINEMENT EXPERIMENT BY USING DOMINANT BEHAVIORS. CLIPS WERE SEGMENTED THE SAME WAY AS ABOVE

Games	#Dominant	#Non-dominant	#Related in the dominant	$P'_{uniform}$
Ballplaying	11	9	6	54.5%
Cooking	20	0	10	50%
Reading	11	9	6	54.5%

Considered in this way, we experiment on the three activity query and use the label information as a clue to get rid of the undominant behavior clips, the final result shows in table III. Where the $P'_{uniform}$ is defined as $P'_{uniform}$ = (#Related in the dominate)/(#Dominant). We could see from the table III. easily that the overall performance of the retrieval has been improved. And in the precision-recall curve Figure 13., the precision improves a little when recall is at 10%-30% .

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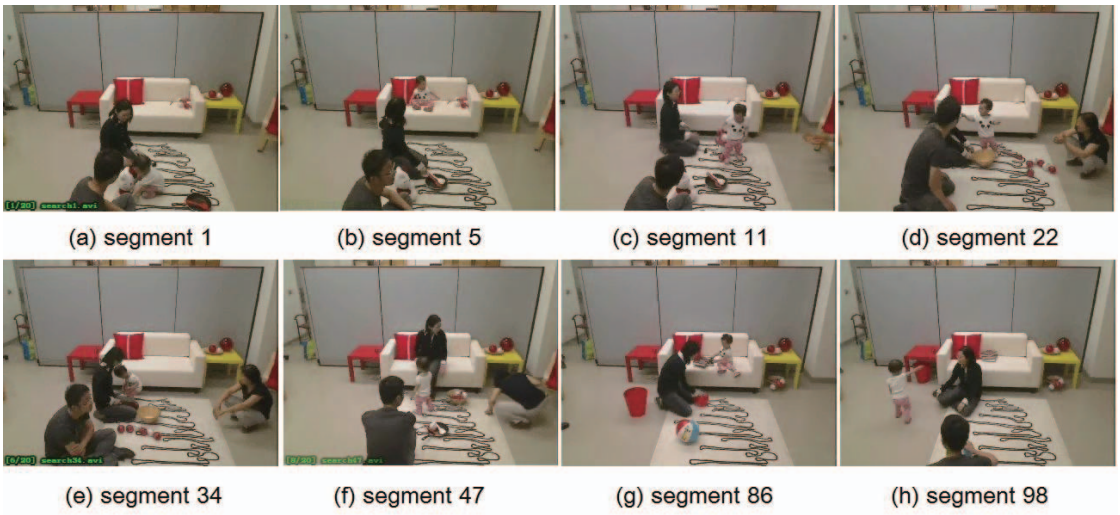

Fig. 10. Snapshots of the unrelated video snapshots

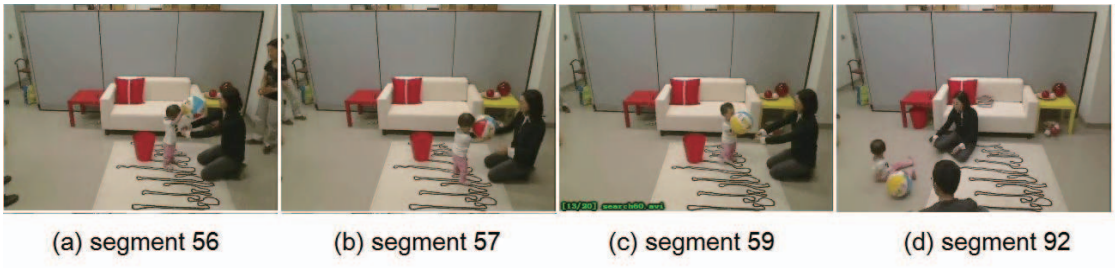

Fig. 11. Snapshots of the related video

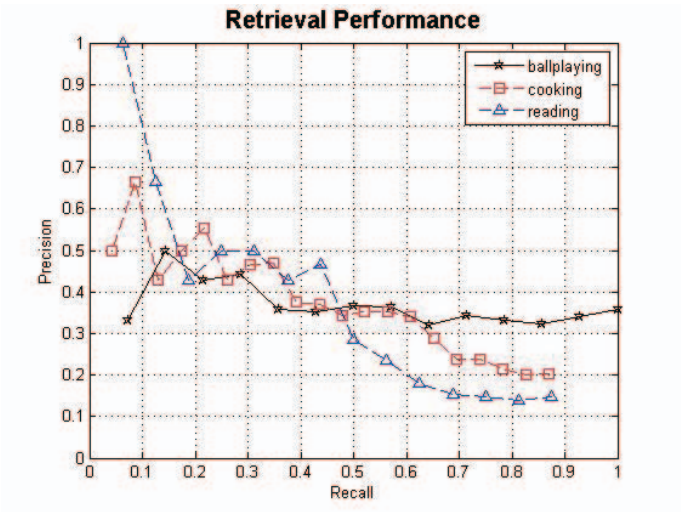

Fig. 13. The retrieval performance of the refinement experiment by using dominant behaviors

D. Context Modeling with Object Recognition

In this stage, we try to find out the object that the kid's playing with. And use this information as a clue to infer the activity. In order to get a fine result of the retrieval, we need to get rid of some unrelated ones that was found out in previous stage. For this purpose, we applied object recognition by SIFT feature matching procedure. And the result shows its improvement in the final retrieval. We use an image patch which contains the object (for example, a ball) as a query input, and compute the SIFT keypoints in the image patch, then we compute the SIFT matches between the input patch and each frame in the video, only the match that counts more than a threshold is to be considered, and the threshold is given by hand. For most of the experiment, the object is not so large to extract enough feature points, the average number is about 10-20. As a result, the number of most computed matches is 3-5, while there are always some false matches, we count the number of the frames that have matches more than the threshold. If the number of the matched frames is larger than a certain number, we consider it to contain the query object. Figure 14 shows the matching result of the object with the scene.

While the sift-based object detection can help us get rid of some unrelated results, and it can improve the retrieval performance in some extent. But sometimes, it still has many false judgement due to its false matching between the object and the scene. Thus, we need use some other robust recognition algorithm for the context modeling in the future.

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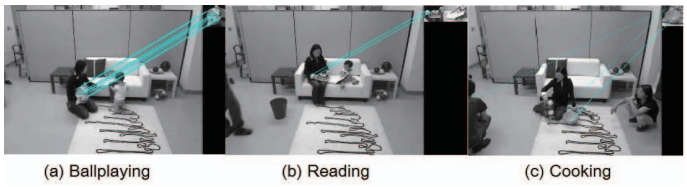

Fig. 14. The sift matching results

V. Conclusion

We present a video retrieval method based on spatio-temporal features matching, which computes the similarity between video segments, and adopts a rank scheme to measure the the degree of similarity between query video and the clips from the dataset. Also, we propose a method to learn dominant child-adult interactive behaviors based on a sequence of home video, and use this to help us get rid of some unrelated query results. What's more, we adopt a context modeling concept to get a refinement of the retrieval results.

Future work includes extending the dataset on a more general one, and the extended dataset may contain many other complex activities. Also, we can take some other activities as the input, or we could make the input query to be performed by other people. Within the matching stage, we would take the spatio and temporal information among the feature points into consideration in the future, thus the matching performance may be improved significantly. In the context modeling stage, we could try some other methods to find out the object, or use another clue to infer the activity.

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